

Exercise

In this exercise, you will create a Quarto document that does the following things. Upload to your GitHub page and submit the resulting URL.

Again read in the CSV at `manctsui.github.io/datasets/timesData.csv` of [Times Higher Education World University Ranking](#). This file was taken from [Kaggle](#).

Problem 1A

Find the three countries with the most universities on the above rankings.

Display this information as a table (`reactable` , `DT` , etc.) with two columns:

- The first column is the country name
- The second column is the number of times the country's universities featured in the above rankings.

Problem 1B

Display the result of Problem 1A as a barchart in `ggplot` or `plotly`.

Problem 2A

Display the top 50 universities in the year 2011 as a table with only two columns:

- First column is the rank.
- Second column is the university name.

Problem 2B

If we are to turn the data frame that results from Problem 1A as a `SharedData` object, would `$rank` column be a good candidate to use as a key? Explain.

Problem 2C

Create a filter slider from `crosstalk` that allows us to narrow our selection of universities based on the rank of the university.

Problem 3

Do two separate scatterplots from our university world ranking dataset restricted to the year 2016:

- First scatterplot is number of students vs. student-staff ratio.
- Second scatterplot is student-staff ratio vs. teaching score.
- Link the two scatterplots together via `crosstalk`.

Problem 4

Now that you finished the above problems, please upload your webpage of solutions rendered from your `qmd` file to your GitHub page and submit the resulting URL.

Problem 5 (Sign up for SAS)

Before the start of class on September 28th, please:

- Create an account with SAS at the link <https://www.sas.com/profile/ui/#/create>
- Check that you can sign into their SAS OnDemand for Academics (formerly SAS Studio) web application at <https://welcome.oda.sas.com/> and eventually get to the following screen with a giant "SAS Studio" and a giant "Launch" button:

 **Launch**

- [Learn](#) to use this client.
- [Clear my saved tabs](#)

